

Phishing Email Analysis Report

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Executive Summary

Phishing pot is a github depository for email analysis containing mails from 2022-2024 for security training purposes.

This report shows the uncovering of malware placed in emails, crypto scams, credential harvesting, etc., which can lead to serious security threats and potential harm to individuals receiving these messages.

This was carried out by analyzing the email header, email body, and technical indicators using various vulnerability assessment tools such as VirusTotal, among others, and by performing a risk assessment followed by a mitigation plan.

Tools Used

VirusTotal

AbuseIPDB

Symantec Bluecoat

sitecheck.sucuri.net

Phishing Email

Phishing is a form of social engineering attack in which a threat actor deceives individuals into revealing sensitive information, often under the guise of a trusted entity or familiar service. The goal is to exploit this information for malicious purposes, including unauthorized access to systems, identity theft, financial fraud.

Threat actors don't stop at simply impersonating others. In phishing emails, they often embed Indicators of Compromise (IOCs) clues that signify malicious intent, this sometimes come as a matter of urgency. One of the most common IOCs is the **email itself**, which can be used as a tool to launch various types of attacks, including but not limited to:

1. Collection of Login Credentials

Phishing emails may contain fake login pages mimicking legitimate services like Microsoft 365, Google, Crypto login page or banking platforms. Victims who enter their credentials unknowingly hand over access to their accounts, enabling attackers to:

- Access emails and cloud files
- Launch further phishing attacks from the victim's account
- Conduct corporate espionage or financial theft

2. Collection of Personal or Corporate Data

Some phishing emails are designed to extract sensitive personal information (like BVN Numbers or bank details) or corporate data (such as an urgency to login into the databases or confidential reports). This data can then be:

- Sold on the dark web
- Used in future attacks (e.g., spear phishing)
- Exploited for blackmail or extortion

3. Deployment of Malware

Phishing emails often carry malicious attachments (e.g., PDFs, Word documents, ZIP files) or links that, when opened, install malware on the recipient's device. This malware may be:

- Ransomware to encrypt data and demand payment

- Keyloggers to capture typed information
- Remote access trojans (RATs) to take over the system

4. Redirection via Authentic-Looking Links

Threat actors may craft links that appear legitimate at first glance (e.g., using typo squatting like `micros0ft.com` instead of `microsoft.com`). When clicked, these links redirect users to:

- Fake login pages
- Malware-hosting sites

The Mail

The mail used as a case study here was sent to the user phishing@pot.org on Thursday, 23rd of May 2024 at 12:19pm by ex.pharmacy.online4@outlook.com

The body of the mail shows a discount on some prescription only medications that are banned in most part of the world.

This is a social engineering method for the threat actor to make a user do their bidding.

Most cases mails like this always come as matter of urgency.

The mail has an embedded URL that redirect the user to another forum.

The sender addressed the mail to reply to the same email address.

The screenshot shows an email interface with a dark header. The sender is 'Viagra/Cialis-0,36\$/0,54\$' with email 'ex.pharmacy.online4@outlook.com'. The subject is 'To Undisclosed recipients;'. The body text says 'Claim your 5% discount code now.' Below this is a large white box with a blue header that reads 'Exclusive 5-10% Discount'. Inside this box is a table listing medications and their prices.

Medication	Price
Cialis	\$0.54
Viagra	\$0.36
Levitra	\$0.99
Female Viagra	\$0.76
Antibiotics/Anti-Depressants/ Allergies/Acne/Asthma/Anti-Acidity/ Anti-Fungus	And more...

Email Metadata Analysis

The sender information:

- Return path vs from: The email has no return path, but it was sent from SA1P223MB0630.NAMP223.PROD.OUTLOOK.COM (::1) after being routed over different servers. This indicates spoofing.
- SPF Record: the SPF record is set to failed, if the domain used discourages the senders ip address as a permitted sender. This means no authentication is present.
- Sender IP address: From scan on AbuseIPDB the Ip address 188.118.210.5 appears to be reported 124 times from 64bdistinct location in the last 1 year, this appears to be an Ip address used for phishing email and Email spam.
- DKIM/DMARC: not configured and ineffective

Conclusion: The email header reveals several IOCs, such as a suspicious domain and failed SPF/DKIM checks, indicating poor domain security and confirming it's a phishing attempt.

The screenshot shows the AbuseIPDB website interface. At the top, there is a navigation bar with the AbuseIPDB logo, a 'LOGIN' button, and a 'SIGN UP' button. Below the navigation bar, there is a search bar with the text 'AbuseIPDB » 188.118.210.5'. The search bar has a green background and a 'CHECK' button. Below the search bar, there is a feedback button on the left. The main content area shows the search results for the IP address 188.118.210.5. It states '188.118.210.5 was found in our database!' and 'This IP was reported 124 times. Confidence of Abuse is 0%: ?'. There is a progress bar showing 0%. Below this, there is a table with the following information:

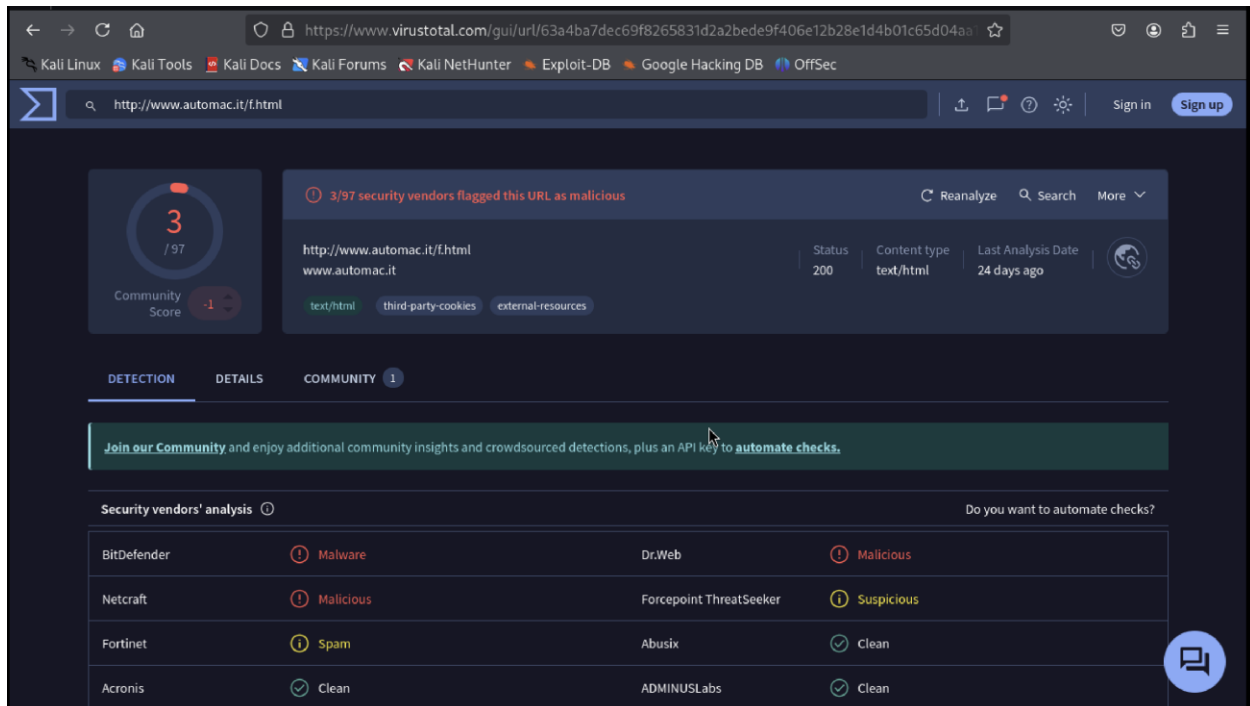
ISP	Hutchison Drei Austria GmbH
Usage Type	Fixed Line ISP
ASN	AS8437
Hostname(s)	sil210005.unixsecurity.at

URL Analysis

The URL `http://www.automac.it/f.html` was tested using the tools listed above.

It was finally redirected to `http://bestlove.sbs/?cp=j9f0hmx9`

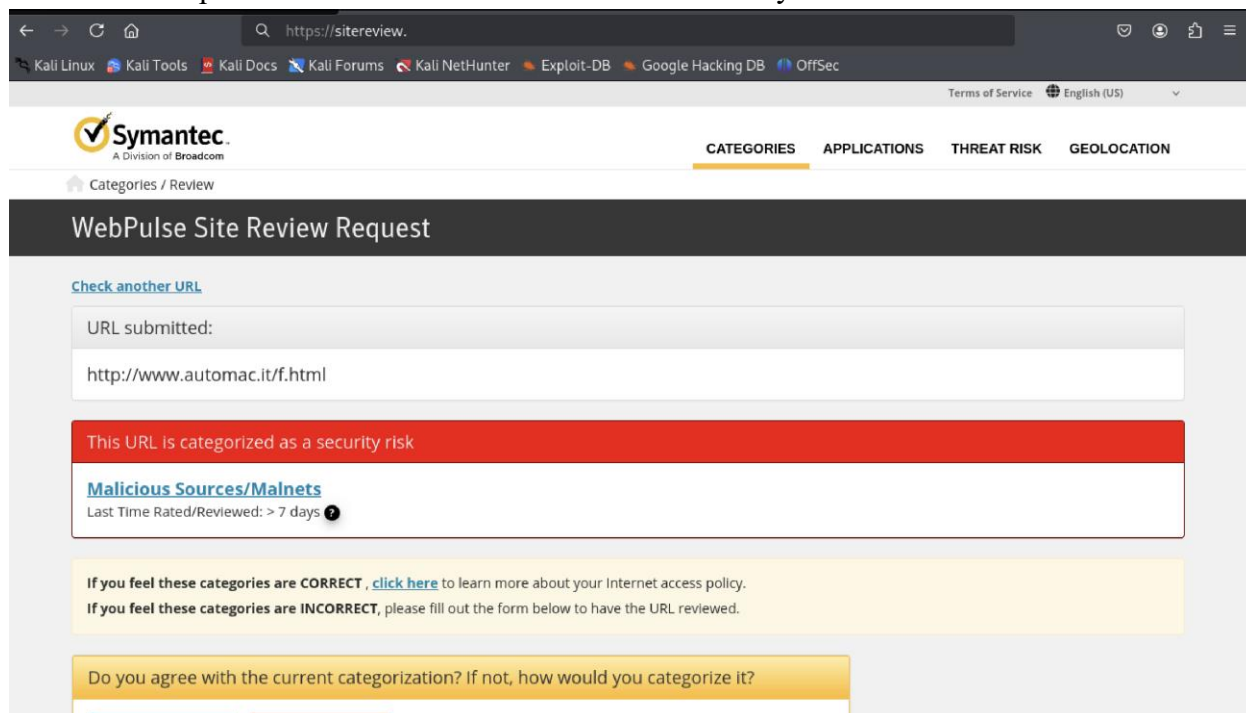
VirusTotal: It is classified as a malicious URL with a community score of 3/97. Three security vendors classified it as a malware, one classified it as spam while another classified it as suspicious.



The screenshot shows the VirusTotal web interface for the URL `http://www.automac.it/f.html`. The browser's address bar shows the full URL. The page header includes navigation links for Kali Linux, Kali Tools, Kali Docs, Kali Forums, Kali NetHunter, Exploit-DB, Google Hacking DB, and OffSec. The main content area displays a community score of 3/97, indicating that 3 out of 97 security vendors flagged the URL as malicious. Below this, there are tabs for DETECTION, DETAILS, and COMMUNITY (1). A banner encourages joining the community for more insights and automated checks. The 'Security vendors' analysis' section shows a table of results from various vendors.

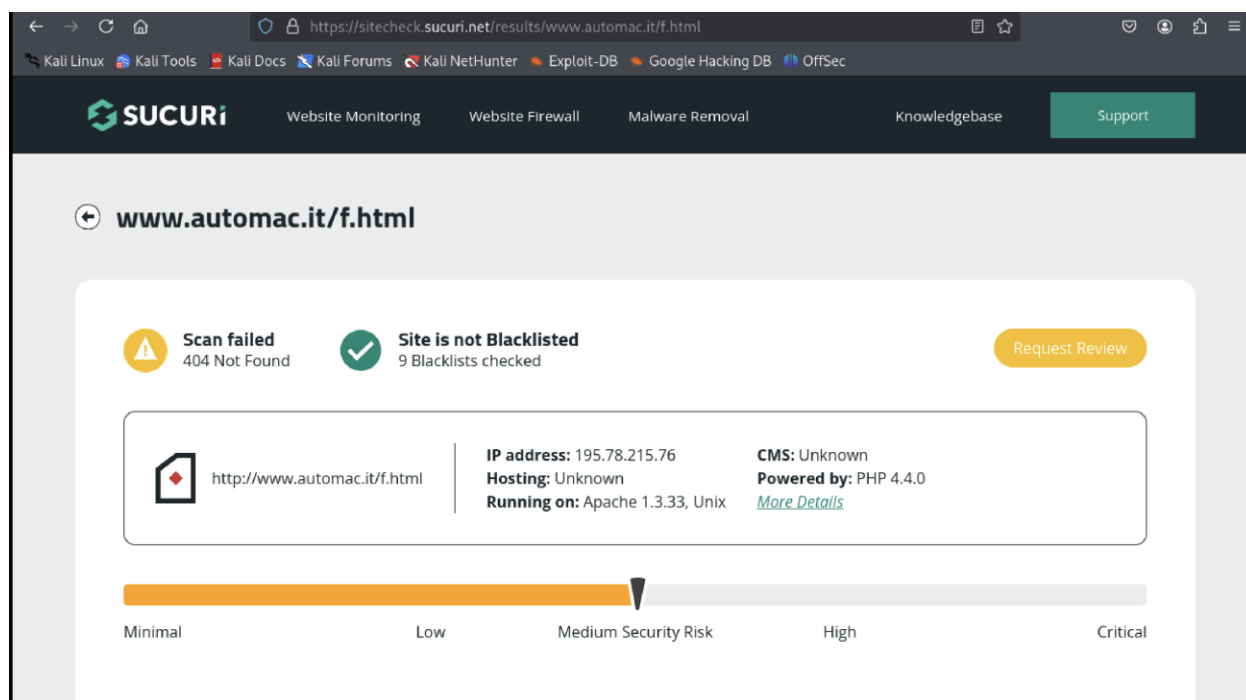
Security vendors' analysis		Do you want to automate checks?	
BitDefender	Malware	Dr.Web	Malicious
Netcraft	Malicious	Forcepoint ThreatSeeker	Suspicious
Fortinet	Spam	Abusix	Clean
Acronis	Clean	ADMINUSLabs	Clean

Symantec Bluecoat: It was classified as a Malnet; this is site that host malware or host purpose of existence is as part of a malware network or a malware ecosystem.



The screenshot shows the Symantec WebPulse Site Review Request page. The browser address bar shows the URL `https://sitereview.`. The page header includes the Symantec logo and navigation links for CATEGORIES, APPLICATIONS, THREAT RISK, and GEOLOCATION. The main heading is "WebPulse Site Review Request". Below this, there is a link "Check another URL". A text box labeled "URL submitted:" contains the URL `http://www.automac.it/f.html`. A red banner indicates "This URL is categorized as a security risk". Below this, a box labeled "Malicious Sources/Malnets" shows "Last Time Rated/Reviewed: > 7 days". A yellow box contains instructions: "If you feel these categories are CORRECT, click here to learn more about your Internet access policy. If you feel these categories are INCORRECT, please fill out the form below to have the URL reviewed." At the bottom, a yellow box asks "Do you agree with the current categorization? If not, how would you categorize it?" with a progress bar below it.

Sitecheck.sucuri.net: It failed on this site and classified it as a medium security risk with a provided Ip address 195.78.215.76 for the URL.



The screenshot shows the Sitecheck.sucuri.net results page for the URL `http://www.automac.it/f.html`. The browser address bar shows the URL `https://sitecheck.sucuri.net/results/www.automac.it/f.html`. The page header includes the SUCURI logo and navigation links for Website Monitoring, Website Firewall, Malware Removal, Knowledgebase, and Support. The main heading is "www.automac.it/f.html". Below this, there are two status boxes: "Scan failed 404 Not Found" and "Site is not Blacklisted 9 Blacklists checked". A "Request Review" button is visible. A box contains the URL `http://www.automac.it/f.html` and technical details: "IP address: 195.78.215.76", "Hosting: Unknown", "Running on: Apache 1.3.33, Unix", "CMS: Unknown", and "Powered by: PHP 4.4.0". A "More Details" link is also present. At the bottom, a security risk scale is shown, ranging from Minimal to Critical, with a marker indicating a "Medium Security Risk".

Looking up the IP address provided by the sitecheck.sucuori. on AbuseIPDB it was found to be reported 10 times

The screenshot shows the AbuseIPDB website interface. At the top, there's a navigation bar with the AbuseIPDB logo, a 'LOGIN' button, and a 'SIGN UP' button. Below the navigation bar, there's a search bar with the text 'Check an IP Address, Domain Name, or Subnet' and a 'CHECK' button. The search results show that the IP 195.78.215.76 was found in the database. It also displays the number of reports (10) and the confidence of abuse (2%). A progress bar indicates the 2% confidence level. Below this, there's a table with details about the IP, including the ISP (IRIDEOS S.P.A.), Usage Type (Data Center/Web Hosting/Transit), ASN (AS3302), and Domain Name (retelit.it).

AbuseIPDB » 195.78.215.76

Check an IP Address, Domain Name, or Subnet
e.g. 105.113.12.229, microsoft.com, or 5.188.10.0/24

195.78.215.76

195.78.215.76 was found in our database!

This IP was reported **10** times. Confidence of Abuse is **2%**: ?

2%

ISP	IRIDEOS S.P.A.
Usage Type	Data Center/Web Hosting/Transit
ASN	AS3302
Domain Name	retelit.it

Key Indicator of Phishing

- **Senders Address:** The sender address doesn't look legitimate enough, it looks like a copy of the original pharmacy address. The email ex.pharmacy.online4@outlook.com comes with an “ex” in the begging and a “4” in the middle of the email address.
- **Attached links:** The mail is embedded with a malicious link when you try to visit the website, the other button on the website prompts you to write an email back to the reply-to attached to the mail.
- **Poor grammar:** Under the pricing in the mail, the last price for the medications was “Add more”, which was addressed to various types of antibiotics. This is a shows the authenticity of the mail to be false.
- **Mismatched URLs:** Hovering over the buttons in the mail shows a different URL from what was displayed.
- **Email Header Anomalies:** The IP address in the email header wasn't a private or static IP address that should prove the anticity of the sender. The mail routed through various unknow severs before getting to the received.

Recommendations

1. Strengthen Email Security Controls

- Implement advanced email filtering solutions to detect and block suspicious emails before they reach users.
- Enforce proper email authentication protocols such as SPF, DKIM, and DMARC to reduce spoofing attempts.

2. Conduct Regular Security Awareness Training

- Train employees on how to identify phishing emails (e.g., suspicious domains, poor grammar, mismatched URLs).
- Encourage reporting of suspicious emails using a clear internal reporting process.

3. Implement URL and Attachment Scanning Tools

- Use automated tools to scan embedded URLs and attachments for potential malware or redirects.

Conclusion

The analysis of the phishing email highlights the presence of multiple Indicators of Compromise (IOCs), including a spoofed sender domain, failed SPF/DKIM checks, malicious URLs, and suspicious email header anomalies. These findings confirm the email as a phishing attempt aimed at exploiting users through social engineering and malware distribution.

To mitigate such threats, the organization must adopt robust email security measures, enforce authentication standards, and strengthen user awareness. A proactive security posture—supported by continuous monitoring and timely response will significantly reduce the likelihood of successful phishing attacks.